

VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



Machine Learning (CO3117)

Assignment

Aerial Scene Classification

Advisor(s): Dr. Nguyễn Đức Dũng

Student(s): Lê Minh Hào

2310846

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1 Introduction

1.1 Motivation

Nowadays, aerial images allow us to measure the Earth’s surface with detailed structures, serving as a significant data source for Earth observation. Aerial scene classification, which aims to automatically label an aerial image with a specific semantic category, is a fundamental problem in understanding high-resolution remote sensing imagery. I believe it is an interesting field in which to apply the knowledge from my machine learning course. Furthermore, it has a wide range of real-world applications, including natural hazard detection, urban planning, disaster management, environmental monitoring, and the control of unmanned aerial vehicles (UAVs). Consequently, aerial scene classification is a suitable problem for studying and gaining practical experience.

1.2 Overview of the dataset

The Aerial Image Dataset (AID) is a large-scale dataset for aerial scene classification. It addresses the limitations of older datasets like UC-Merced and WHU-RS19, which are relatively small and have reached performance saturation. AID is constructed by collecting 10,000 sample images from Google Earth imagery.

The dataset consists of 30 distinct aerial scene types, ranging from natural landscapes (e.g., forest, river, mountain) to man-made structures (e.g., airport, dense residential, stadium). The images in AID are multi-source, as Google Earth aggregates data from various remote imaging sensors. Moreover, the samples are carefully chosen from different regions around the world—including China, the United States, and Europe—and are extracted at different times and seasons. These diverse imaging conditions introduce greater challenges for scene classification. Some samples of each class are shown in Fig. 1.1.

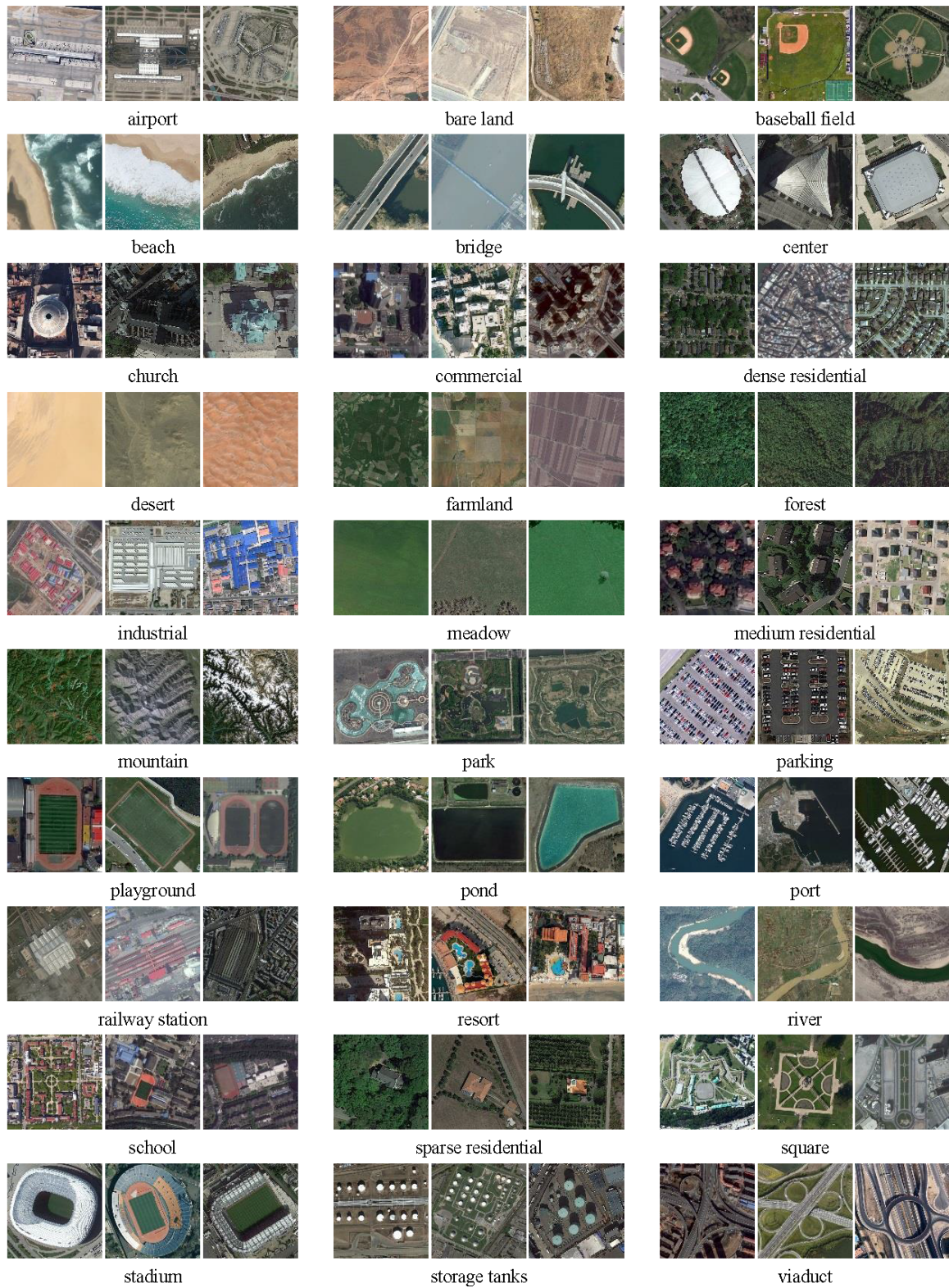


Figure 1.1: Samples of the AID dataset^[1]

2 Approach

To address the challenge of Aerial Scene Classification, I utilized a classical machine learning framework: a Support Vector Machine (SVM) equipped with a Radial Basis Function (RBF) kernel. This classifier operates on a fused feature set comprising Histogram of Oriented Gradients (HOG) and Color Histograms. Feature fusion is particularly suitable for this domain, as it allows the model to leverage both structural information and color distribution. Furthermore, SVMs are mathematically robust in high-dimensional spaces—a common characteristic of fused feature vectors—where other algorithms, such as Decision Trees, are prone to overfitting.

2.1 Support Vector Machine (SVM)

2.1.1 Formulation

Support Vector Machines are supervised learning models used for classification and regression analysis. In a binary classification setting with linearly separable data, the SVM seeks to find the optimal hyperplane that separates the classes with the maximum margin.

Given a dataset of N training samples $\{(x_1, y_1), \dots, (x_N, y_N)\}$, where $x_i \in \mathbb{R}^d$ represents the feature vector and $y_i \in \{-1, 1\}$ represents the class label, the SVM solves the following optimization problem to minimize the weight vector w :

$$\min_{w,b} \frac{1}{2} \|w\|^2 \quad (2.1)$$

Subject to the constraints:

$$y_i(w^T x_i + b) \geq 1, \quad \forall i = 1, \dots, N \quad (2.2)$$

2.1.2 Kernel Support Vector Machine

Aerial scene data is rarely linearly separable in its original feature space. To handle this non-linearity, we employ the "Kernel Trick." This method implicitly maps input vectors into a higher-dimensional feature space via a mapping function $\phi(x)$. The SVM algorithm is then applied in this new space. The kernel function K computes the dot product in the high-dimensional space without explicitly calculating the transformation:

$$K(x_i, x_j) = \phi(x_i)^T \phi(x_j) \quad (2.3)$$

2.1.3 Why RBF Kernel?

For this project, we selected the Radial Basis Function (RBF) kernel. The RBF kernel is defined as:

$$K(x, x') = \exp(-\gamma \|x - x'\|^2) \quad (2.4)$$

where γ defines the influence of a single training example.

We chose the RBF kernel over Linear or Polynomial kernels for two primary reasons:

- **Infinite Dimensional Mapping:** The RBF kernel maps samples into an infinite-dimensional space, providing a high capacity to model complex, non-linear boundaries between the 30 distinct aerial classes (e.g. distinguishing a "Park" from a "Forest").
- **Locality:** RBF is a local kernel, meaning the prediction for a new point is influenced by the distance to support vectors. This is intuitive for image classification, where visually similar images (close in feature space) should belong to the same class.

2.2 Histogram of Oriented Gradients (HOG)

2.2.1 Algorithm Overview

The Histogram of Oriented Gradients (HOG) is a feature descriptor used in computer vision for object detection. It counts occurrences of gradient orientation in localized portions of an image. The implementation follows these steps:

1. **Gradient Calculation:** For every pixel, we calculate the gradient magnitude $m(x, y)$ and orientation $\theta(x, y)$:

$$m(x, y) = \sqrt{g_x^2 + g_y^2}, \quad \theta(x, y) = \arctan\left(\frac{g_y}{g_x}\right) \quad (2.5)$$

where g_x and g_y are the horizontal and vertical gradients.

2. **Cell Histograms:** The image is divided into small spatial regions called "cells" (e.g. 8×8 , 32×32 pixels). A local 1-D histogram of gradient directions is accumulated over the pixels of the cell.
3. **Block Normalization:** To account for changes in illumination and contrast, local groups of cells are accumulated into larger spatial blocks and normalized.

2.2.2 Role in Feature Fusion

In aerial imagery, man-made structures such as "Airports," "Bridges," and "Squares" are defined by rigid shapes and strong edges. HOG is explicitly designed to capture this structural information. By using HOG, our model can distinguish classes based on their geometric layout, regardless of their color.

2.3 Color Histogram

2.3.1 Formulation

While HOG captures shape, it discards chromatic information. To address this, we compute a Color Histogram, which represents the distribution of colors in an image. For an image I , the histogram H for a specific color channel is defined as:

$$H(k) = n_k \quad (2.6)$$

where n_k is the number of pixels with intensity value k . In our approach, we compute histograms for the Red, Green, and Blue channels and concatenate them. We also normalize the histogram to treat it as a probability distribution:

$$p(k) = \frac{H(k)}{\sum_{i=0}^{B-1} H(i)} \quad (2.7)$$

where B is the number of bins (set to 32 in our experiment).

2.3.2 Role in Feature Fusion

Many natural aerial scenes, such as "Forests," "Deserts," and "Oceans," are unstructured and primarily defined by their texture and color palette. For example, HOG might find similar "noisy" gradients in both a forest and a desert, but the Color Histogram will easily distinguish them based on the prevalence of green versus yellow pixels. Fusing this with HOG provides a comprehensive descriptor that captures both shape and material properties.



3 Experiments

3.1 Experimental Setup

The experiments were conducted on a laptop equipped with a Ryzen 7 3750H CPU and 32GB of RAM. Throughout this section, various configurations were evaluated to assess their impact on model performance. These configurations involved adjusting the image size, the training/testing split percentage, the randomization seed, the SVM kernel, and the "pixels per cell" parameter used during HOG feature extraction.

Typically, the pixels per cell value was set to the image size divided by 8 or 16. A larger value results in fewer feature vectors per image, leading to faster training times but potentially lower resolution features. To ensure consistency across all test results, the random seed was fixed at 42. Unless otherwise noted, the training set comprised 80% of the original dataset.

3.2 Impact of Feature Fusion

Initially, the HOG transform was the sole feature descriptor used for SVM training; however, the performance was suboptimal and did not meet expectations. I observed that for certain scene types—such as deserts, beaches, and forests—color plays a pivotal role in identification. Consequently, we incorporated color histograms into the feature fusion process. As illustrated in Table 3.1, this addition significantly improved the results.

Table 3.1: Impact of Color Histogram on Accuracy

IMG_SIZE	PIXELS_PER_CELL	Accuracy (No Color)	Accuracy (With Color)
64 x 64	8 x 8	39.65%	54.05%
128 x 128	16 x 16	47.75%	60.75%
256 x 256	32 x 32	52.50%	63.05%

It is evident that the model's performance increased by approximately 10-15%. To understand this improvement in greater detail, we analyzed the F1 score for each class using an image size of 128x128 and a pixels per cell size of 16x16.

Table 3.2: Comparison of F1-Scores per Class: HOG only vs. Feature Fusion (HOG + Color)

Scene Type	F1 (No Color)	F1 (Color)	Δ F1
Airport	0.48	0.57	+0.09
BareLand	0.32	0.76	+0.44
BaseballField	0.67	0.68	+0.01
Beach	0.58	0.75	+0.17
Bridge	0.78	0.78	0.00
Center	0.46	0.57	+0.11
Church	0.27	0.51	+0.24
Commercial	0.29	0.40	+0.11
DenseResidential	0.57	0.61	+0.04
Desert	0.50	0.86	+0.36
Farmland	0.38	0.53	+0.15
Forest	0.60	0.76	+0.16
Industrial	0.29	0.42	+0.13
Meadow	0.56	0.82	+0.26
MediumResidential	0.21	0.48	+0.27
Mountain	0.51	0.68	+0.17
Park	0.43	0.50	+0.07
Parking	0.37	0.61	+0.24
Playground	0.80	0.83	+0.03
Pond	0.79	0.82	+0.03
Port	0.49	0.55	+0.06
RailwayStation	0.44	0.52	+0.08
Resort	0.25	0.25	0.00
River	0.42	0.57	+0.15
School	0.15	0.27	+0.12
SparseResidential	0.49	0.70	+0.21
Square	0.25	0.32	+0.07

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Table 3.2: Comparison of F1-Scores (Continued)

Scene Type	F1 (No Color)	F1 (Color)	Δ F1
Stadium	0.78	0.83	+0.05
StorageTanks	0.22	0.36	+0.14
Viaduct	0.65	0.71	+0.06
Overall Accuracy	47.75%	60.75%	+0.13

As predicted, for scenes where color similarity is a defining characteristic, the accuracy increased noticeably (e.g., Bare Land, Desert, Beach). Overall, this configuration yielded a 13% increase in accuracy across the 30 distinct scene types.

3.3 Impact of Image Resolution and Granularity

Next, we investigated different image sizes and "pixels per cell" setups, as these parameters most significantly impact training results. A larger image size provides more detail, whereas a larger pixels-per-cell value promotes generalization and robustness, as it captures the global structure without relying on exact pixel alignment.

Table 3.3: Performance Comparison: Image Size vs. Pixels Per Cell

IMG SIZE	PIXELS PER CELL	ACCURACY	TRAINING TIME
64 x 64	4 x 4	44.75%	472.47s
64 x 64	8 x 8	54.05%	103.77s
64 x 64	16 x 16	62.55%	16.10s
64 x 64	32 x 32	61.20%	4.87s
128 x 128	8 x 8	48.25%	438.38s
128 x 128	16 x 16	60.75%	93.83s
128 x 128	32 x 32	67.15%	16.02s
128 x 128	64 x 64	63.85%	5.78s
256 x 256	16 x 16	55.05%	407.02s
256 x 256	32 x 32	63.05%	87.74s

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Table 3.3: Performance Comparison (Continued)

IMG SIZE	PIXELS PER CELL	ACCURACY	TRAINING TIME
256 x 256	64 x 64	70.25%	11.43s
256 x 256	128 x 128	63.45%	4.98s
512 x 512	32 x 32	56.95%	393.69s
512 x 512	64 x 64	64.60%	92.63s
512 x 512	128 x 128	71.40%	10.58s
512 x 512	256 x 256	64.70%	4.67s

A trend is observable across all four image sizes: when the ratio between the image size and the pixels per cell decreases by half from 16 (i.e., ratios of 16, 8, 4, 2), the accuracy initially increases, peaking when the ratio is 4, and then decreases at 2. This suggests that for this dataset, overall structural details are more critical than fine pixel textures.

Furthermore, as the ratio decreases, the number of feature vectors decreases, leading to reduced training times. However, the feature extraction time (not recorded in the log) increases significantly due to the higher number of calculations required. Based on these findings, we selected the standard 512x512 image size with 128x128 pixels per cell as the standard for upcoming experiments.

3.4 Kernel Selection

We also tested different SVM kernels to evaluate their performance. These tests were conducted using our standard configuration (512x512 image size).

Table 3.4: Accuracy Comparison of Different SVM Kernels

Kernel	Accuracy
linear	56.90%
poly	57.20%
sigmoid	36.60%
rbf	71.40%

As is well known, the RBF kernel outperforms the others and is suitable for general classification problems where data properties are unknown. Since the data in this case is



hardly linearly separable, the linear kernel is not an optimal choice despite its speed and simplicity. Similarly, both the poly and sigmoid kernels proved to be ineffective for this aerial scene classification task.

3.5 Comparison with State-of-the-Art Methods

Finally, we employed the standard configuration with the RBF SVM kernel to benchmark our results against existing literature. To reduce the influence of randomness and obtain reliable results, we calculated the average accuracy over 100 repetitions. We compared our performance on 20% and 50% training sets against low, mid, and high-level methods presented by Xia et al.^[1] when they introduced this dataset.

According to Xia et al., methods can be categorized as follows:

- **Low-level methods:** These partition an image into small patches and use features like spectral, texture, or structural information (e.g., SIFT, LBP, Color Histogram (CH)).
- **Mid-level methods:** These build a scene representation by encoding low-level local feature descriptors. Our SVM method, utilizing HOG and CH features, falls into this category. We also include the Improved Fisher Kernel (IFK), which uses Gaussian Mixture Models, as a reference for this class.
- **High-level methods:** These utilize deep learning techniques to achieve impressive classification results.

Table 3.5: Comparison with Benchmark Methods from Xia et al.^[1]

Method	Type	20% dataset	50% dataset
SIFT	low-level	13.50 ± 0.67	16.76 ± 0.65
LBP	low-level	26.26 ± 0.52	29.99 ± 0.49
CH	low-level	34.29 ± 0.49	37.28 ± 0.46
Gist	low-level	30.61 ± 0.63	35.07 ± 0.41
IFK(SIFT)	mid-level	71.92 ± 0.41	78.99 ± 0.48
IFK(LBP)	mid-level	60.11 ± 0.56	69.22 ± 0.72
IFK(CH)	mid-level	59.60 ± 0.66	67.49 ± 0.81

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Table 3.5: Comparison with Benchmark Methods (Continued)

Method	Type	20% dataset	50% dataset
Our method (SVM)	mid-level	58.13 ± 0.59	66.81 ± 0.63
CaffeNet	high-level	86.86 ± 0.47	89.53 ± 0.31
VGG-VD-16	high-level	86.59 ± 0.29	89.64 ± 0.36
GoogLeNet	high-level	83.44 ± 0.40	86.39 ± 0.55

From the results in Table 3.5, we can quantify the significant performance gap between the two paradigms. Deep learning approaches, such as VGG-VD-16, achieve accuracy rates nearing 90%, whereas our classic SVM-based approach hovers around 66% (on the 50% dataset split). This represents a substantial difference of approximately 20-25%. The primary reason for this disparity lies in feature extraction. Classic ML approaches rely on hand-crafted descriptors (like HOG and Color Histograms) which are limited by human design and only capture low-level properties. In contrast, Deep Learning models automatically learn hierarchical, high-level semantic features directly from the data, allowing them to distinguish complex aerial scenes with far greater precision. While our work trails behind these modern techniques, it demonstrates competitive results when compared to other mid-level methods introduced in the original paper^[1].



4 Conclusion

Aerial scene classification is a fundamental remote sensing task that labels high-resolution imagery for applications ranging from urban planning to disaster monitoring. As the volume of aerial data expands, developing robust automated systems is critical to efficiently interpret complex surface structures and extract meaningful insights.

Our experiments revealed that relying solely on structural features like HOG is insufficient for this task. However, fusing these with Color Histograms to capture spectral properties significantly boosted performance. Furthermore, careful optimization of the SVM's RBF kernel and feature granularity proved essential for maximizing the potential of classical machine learning methods.

Despite these improvements, our best model achieved a maximum accuracy of 66%, significantly lagging behind state-of-the-art Deep Learning benchmarks which approach 90%. This substantial gap underscores the inherent limitations of hand-crafted features. To achieve superior results, future work must transition to Deep Learning architectures like CNNs, which can automatically learn high-level semantic representations directly from data.

For implementation details, our repository containing the source code and experimental logs is publicly available at

<https://github.com/HaoXiaolin05/Aerial-Scene-Classification.git>.



References

- [1] Gui-Song Xia, Jingwen Hu, Fan Hu, Baoguang Shi, Xiang Bai, Yanfei Zhong, Liangpei Zhang, and Xiaoqiang Lu. Aid: A benchmark data set for performance evaluation of aerial scene classification. *IEEE Transactions on Geoscience and Remote Sensing*, 55(7):3965–3981, 2017.